

Effects of Integrating an AI Learning Companion into a Self-Regulated Learning 4L Framework on Elementary Mathematics Learning

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AI Learning Companions Require Regulatory Scaffolding



① AI and Intelligent Tutoring Systems in Education

Intelligent tutoring systems can support differentiated instruction through learner modeling, adaptive feedback, and learning analytics. (Chen et al., 2020; Labadze et al., 2023; Lin et al., 2023; Silva et al., 2022)

② GenAI as Digital Scaffold

Generative AI may serve as a digital scaffold for self-regulated learning by providing real-time feedback, prompts, and step-by-step guidance. (Chiu, 2024; Wu et al., 2023)

③ Potential Tension

Structured AI guidance may facilitate self-regulated learning, but excessive reliance on AI-generated responses may undermine learner agency. (Jin et al., 2023; Lee & Low, 2024)

④ Research Gap

It remains underexplored whether AI reliance and SRL engagement patterns differ across achievement levels. (Trinovita et al., 2025)

The Scaffold–Dependence Tension in GenAI-Supported Learning



Scaffold

- Real-time feedback
- Step-by-step guidance
- Planning, monitoring, and reflection
- Differentiated support

(Chiu, 2024; Wu et al., 2023)

Dependence

- Passive answer-seeking
- Procedural shortcutting
- Reduced learner agency

(Jin et al., 2023; Lee & Low, 2024)

It remains underexplored whether this scaffold–dependence tension manifests differently across achievement levels. This study addresses this gap by comparing TALPer-mediated regulatory engagement between high- and low-achieving learners.

Aligning SRL 4L with Zimmerman's SRL Cycle

Zimmerman (2002) SRL Cycle

Forethought

Goal Setting &
Prior-Knowledge Activation

Performance

Monitoring, Strategy Use &
Peer Adjustment

Self-Reflection

Error Noticing &
Conceptual Consolidation

Ho (2014) SRL 4L Framework

Self-learning

Goal Setting &
Prior-Knowledge Activation

Co-learning

Monitoring &
Scaffolded Feedback

Mutual learning

Cross-Group Strategy Sharing

Teacher-guided Learning

Misconception Correction &
Conceptual Integration

The SRL 4L phases map onto Zimmerman's cycle. This mapping informed the design of the SRL coding scheme and ENA dimensions adopted in the present study.

Taiwan Adaptive Learning Platform (TALP, developed by the MOE and NTCU since 2017)



TALP

Accounts

4M⁺

Students

220K⁺

Teachers

4235

Schools



Updated on June 12, 2026



adl.edu.tw

TALPer: The TALP AI Learning Companion (based on ChatGPT and Gemini)



T-TALPer

AI Assistant
for Teachers

TALPer was launched nationwide in
September 2024 and has reached
approximately 700,000 users.



Domain Specific
(S-TALPer)

AI Tutor
for Students

Domain General
(G-TALPer)

AI學伴互動討論區

e度: Hi! 小柚! 我是AI學伴e度, 讓我們一起來學習「三角形的全等性質與相似性質」這個單元吧! 請嘗試操作右邊畫面中間的兩個三角形, 試試看能否將兩個三角形完全疊合, 並將結果填列在作答區的表格中。

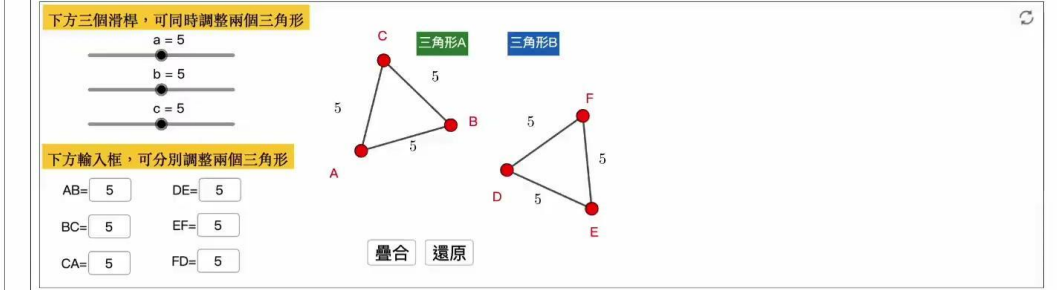
可以再講一次嗎? 為什麼呢?
這一題需要用到哪些概念? 可以出一題類似題給我嗎?

如果任何問題, 可以將問題輸入這裡提問喔!

送出

三角形的全等性質與相似性質

請嘗試操作下方的三角形, 試試看能否將兩個三角形完全疊合, 並將結果填列在作答區的表格中。



作答區

AB	BC	CA	DE	EF	FD	是否可以完全疊合 (請回答是或否)
5	6	5	5	5	6	
3	4	5	6	8	10	
4	7	9	4	7	9	
6	6	6	6	6	6	

送出答案

Lin, T.-C., Shih, Y.-T., & Li, C.-H. (2026). Designing and Evaluating a 5E-Structured GenAI Coach for Guided Inquiry: A Pedagogy-to-Prompt Engineering Framework. *Education Sciences*, 16(3), 384. <https://doi.org/10.3390/educsci16030384>

Kuo, B. C., Bai, Z. E., & Lin, C. H. (2026). Developing an AI Learning Companion for Mathematics Problem Solving in Elementary Schools. *Computers & Education*, 240, 105463.

Research Questions



RQ1

Does TALPer-integrated SRL 4L improve learning outcomes compared with SRL 4L alone?

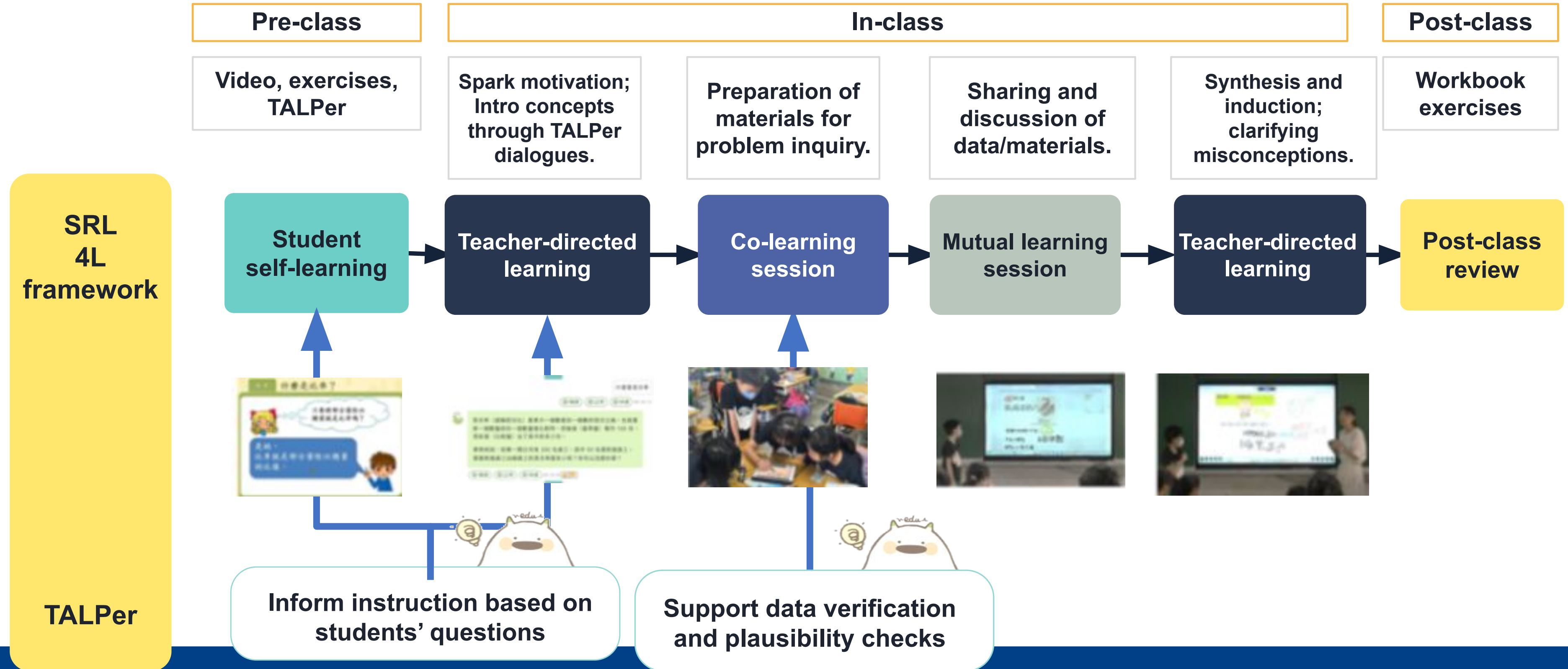
RQ2

Are there any differences in the application of SRL 4L between high- and low-achieving students?

RQ3

What differences in SRL cognitive network structures are revealed by ENA maps between high- and low-achieving students?

Pedagogical Framework: Integrating TALPer into SRL 4L



Research Design and Participants

44

fifth-grade students

Public elementary school, Taiwan

5

lessons on Ratios & Percentages

Delivered over 2 weeks

Comparison of the Two Groups

Experimental Group (n = 21)

- ▶ SRL 4L instruction with TALPer support throughout
- ▶ Pre-class: TALP videos and TALPer dialogues
- ▶ In-class: teachers leveraged TALPer logs
- ▶ TALPer was integrated across Co-, Mutual-, and Teacher-guided learning phases

Control Group (n = 23)

- ▶ Identical SRL 4L instructional structure
- ▶ Only difference: no TALPer access
- ▶ Pre-test scores: no significant group difference ($p = .376$)
- ▶ TALPer integration can thus be treated as the primary treatment condition.

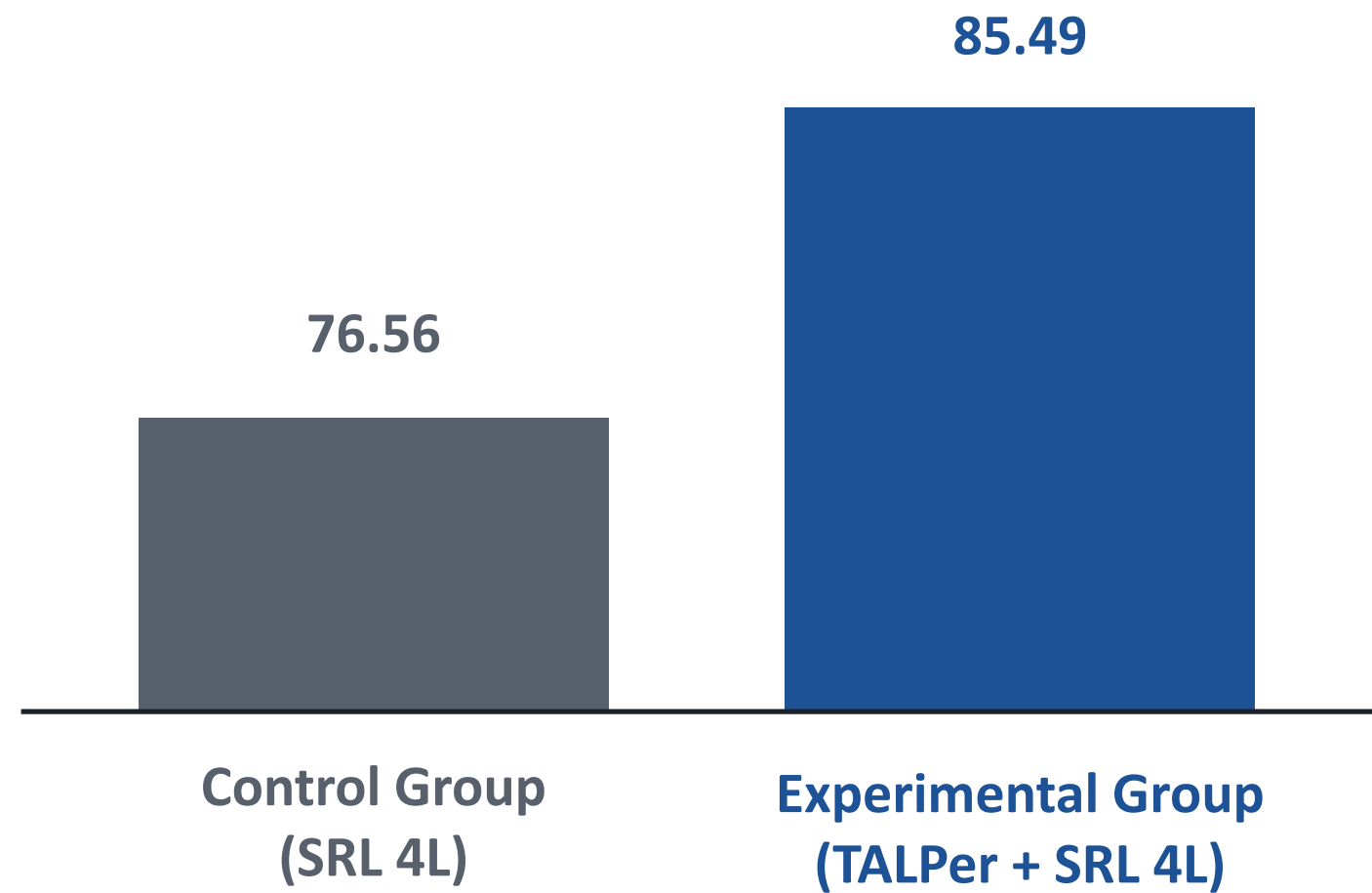
SRL Coding Framework and ENA Dimensions

Main category	Sub-category	Code	Explanation
Metacognition (M)	Setting Goals	M.SG	Proactively sets or adjusts learning objectives or sequences.
	Selecting Strategies	M.SS	Selects or modifies learning strategies, modalities, or task forms.
	Monitoring & Evaluation	M.ME	Real-time awareness and evaluation of conceptual understanding.
	Reflection	M.R	Reviews problem solving to identify errors or revise strategies.
Cognition (C)	Conceptual Understanding: Ask	C.CUA	Asks questions about concepts, principles, meanings, or applications.
	Conceptual Understanding: Explain	C.CUE	Articulates or constructs conceptual understanding.
	Procedural Knowledge: Ask	C.PKA	Asks about computational methods or problem solving steps.
	Procedural Knowledge: Execute	C.PKE	Performs actual computations or formulates equations/answers.

ENA maps code co-occurrences into a 2D space. In this model, the X-axis is interpreted as relative cognitive tendencies, whereas the Y-axis reflects differences in network complexity and

Results: TALPer-Supported SRL 4L Improved Post-Test Performance

Adjusted post-test means



ANCOVA result

$$F(1,41) = 4.71$$

$$p = .036$$

$$\eta^2 = .114$$

Pre-post gains: Control +14.45; TALPer +26.45 (both $p < .001$)

Results: TALPer Usage Patterns Differed Significantly by Achievement Level

High-achieving students

- Goal setting
- Conceptual questions
- Monitoring and reflection
- Iterative strategy use

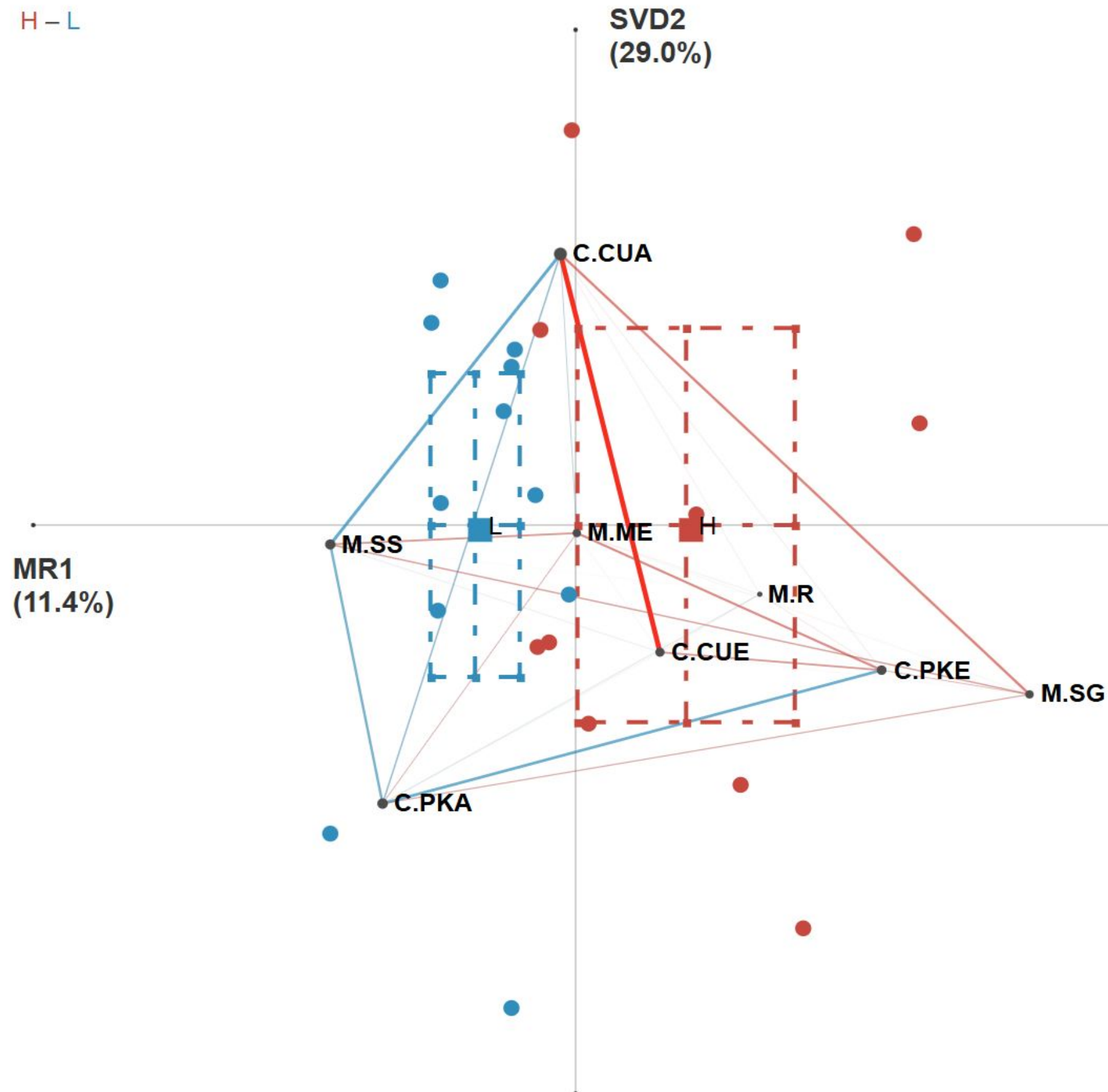
Low-achieving students

- Procedural strategy selection
- Procedural asking
- Answer-seeking without reflection
- Minimal self-monitoring

$t(12.07) = 4.05, p < .001, \text{Cohen's } d = 1.83$

Experimental group only: high-achievers (n = 10) vs. low-achievers (n = 11)
Comparison: ENA X-axis centroid difference between ability groups

Results: ENA Revealed Integrated vs. Fragmented Regulatory Networks



**High-achieving students:
integrated, multidirectional
regulatory network**

**Low-achieving students:
fragmented, procedural,
ask-execute pattern**

Discussion: Findings in Relation to Prior Literature

AI Support, Learner Agency, and Regulatory Awareness

Prior Literature



GenAI can support SRL processes (Chiu, 2024; Jin et al., 2023)

- Provides feedback and prompts
- Supports planning, monitoring, and reflection
- Facilitates self-regulated learning



Effective AI use requires learner agency (Lee & Low, 2024)

- Passive reliance on AI may reduce learner agency
- Critical engagement with AI remains essential

This Study Adds



TALPer + SRL 4L significantly improves mathematics learning

Students supported by TALPer achieved significantly higher learning outcomes.



TALPer effectiveness depends on regulatory awareness

Not all learners benefited from AI in the same way. The critical differentiating factor was learners' capacity for self-regulation and reflective engagement.



AI can support learning, but its effectiveness depends on how learners regulate their engagement.

Discussion: Findings in Relation to Prior Literature

ENA, Learning Pathways, and Cognitive Network Structures

Prior Literature



ENA reveals meaningful cognitive structures

(Shaffer et al., 2016; Reid et al., 2025)

- ENA uncovers connections among cognitive processes
- Network structures can provide evidence of differences in learning processes

This Study Adds



Different AI-supported learning pathways emerged

High-achieving students:

Goal setting · Conceptual inquiry · Monitoring · Reflection

Low-achieving students:

Procedural asking · Answer seeking · Limited reflection



ENA revealed distinct cognitive network structures

High-achieving students:

Integrated · Multidirectional · Densely connected

Low-achieving students:

Fragmented · Procedural · Less reflective



ENA helps reveal not only whether AI-supported learning differs across learners, but also how these differences are structurally organized.

Acknowledgments & Contact



Thank you. Questions and discussion welcome.

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